

Track 3

Adaptation of Dynamic Cities to Extreme Climatic Conditions

Computer vision-based assessment of building facade vulnerability and debris fall risk zones on the street during seismic events using street-level photography

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ISOCARP
World Planning
Congress
RIYADH | الرياض
1 - 4 DECEMBER 2025

Cities & Regions in Action
Planning Pathways to
Resilience and Quality of Life

Computer vision-based assessment of building facade vulnerability and debris fall risk zones on the street during seismic events using street-level photography

TYPE:

Research

LOCALISATION:

Phnom Penh, Phnom Penh,
Cambodia

POPULATION:

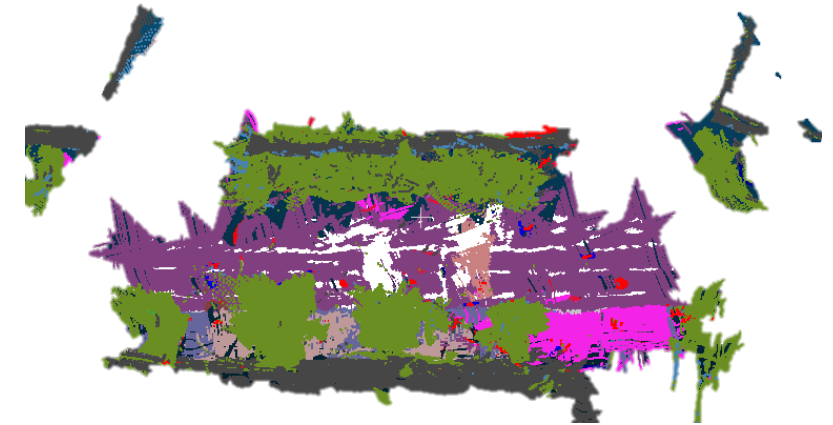
2,352,851

CITY SIZE:

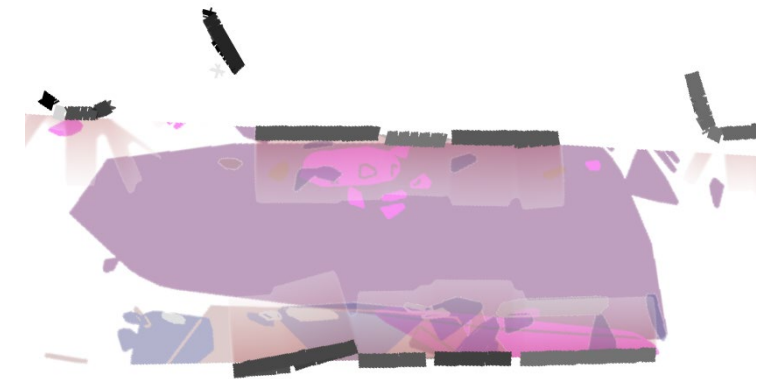
695.46 km²

Key Findings

- Segmentation of the 3d point clouds
- Extracted the building's layout from segmented point clouds
- Estimated risk and safe zones during seismic events



3d Segmented point cloud



Risk zone assessment



Context of the Study - Damages of seismic activities



Before seismic activity (Nation Thailand, n.d)



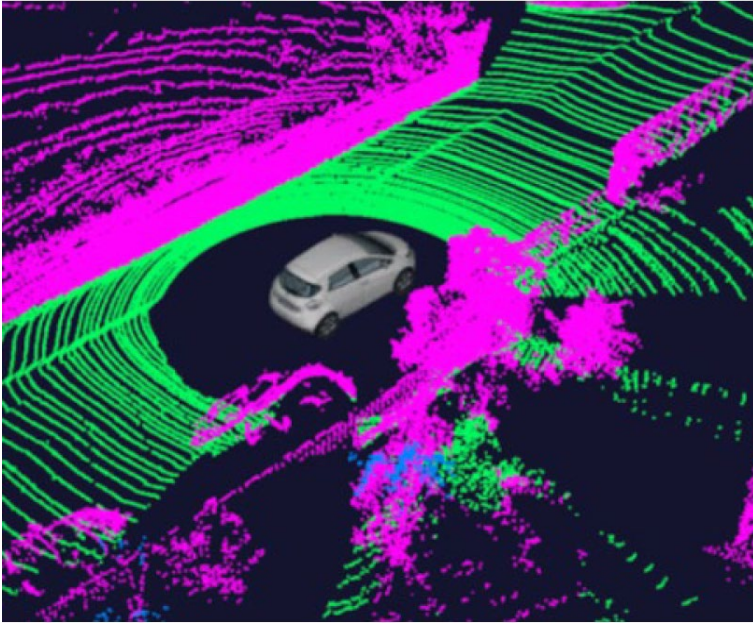
After seismic activity (Khaosod English, 2025)

Nation Thailand (n.d.) [Before seismic activity] [Online image].
Available at: (<https://www.nationthailand.com/news/politics/40048176>) (Accessed: 8 October 2025).
Khaosod English (2025) [After seismic activity] [Online image].

Available at: (<https://www.khaosodenglish.com/news/2025/03/28/tremors-cause-bangkok-building-collapse-trapping-over-80-workers/>) (Accessed: 8 October 2025).

Computer vision-based assessment of building facade vulnerability and debris fall risk zones on the street during seismic events using street-level photography, Leng Monykandy, Gu Zongchao / Southeast University / monykandyleng@gmail.com

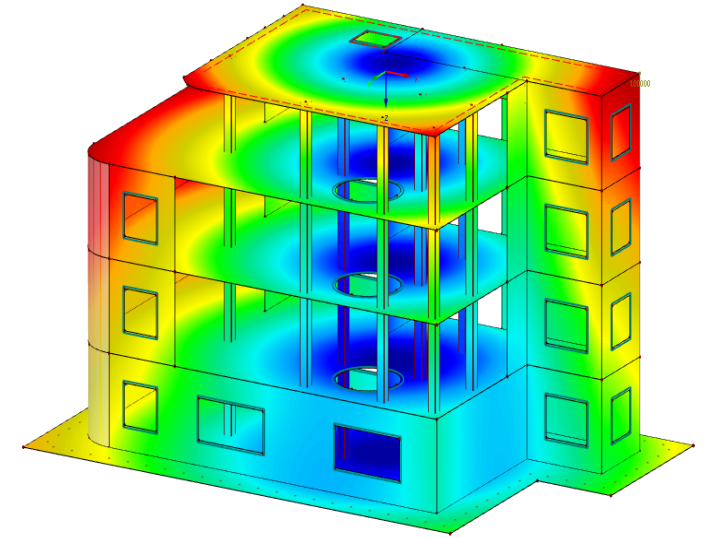
Context of the Study – Existing seismic assessment methods



Lidar Scanning



Drone Photogrammetry

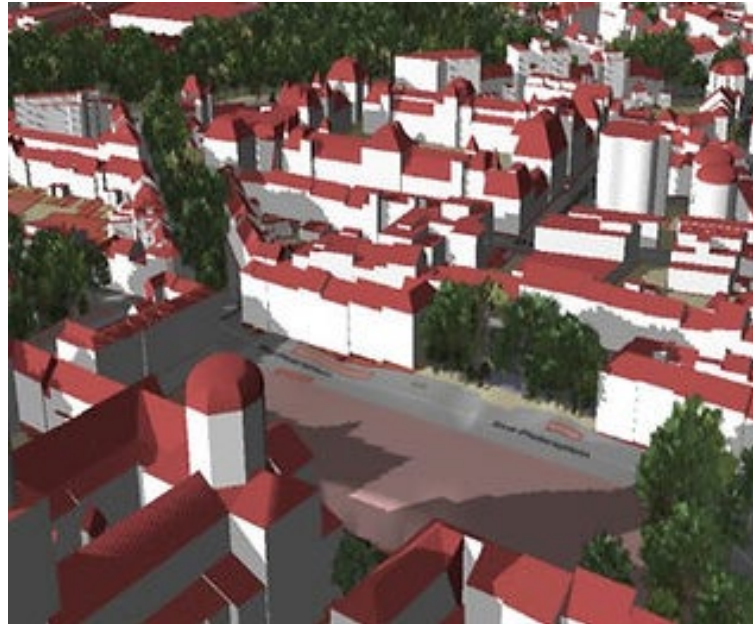


Traditional seismic assessment

Context of the Study – Technological and reality gap



Flanders region, Belgium
Digital Urban Twins (n.d)



DTC of Flanders region, Belgium
Digital Urban Twins (n.d)



Spaces affecting pedestrians
Google Earth (2025)

Google Earth (2024)

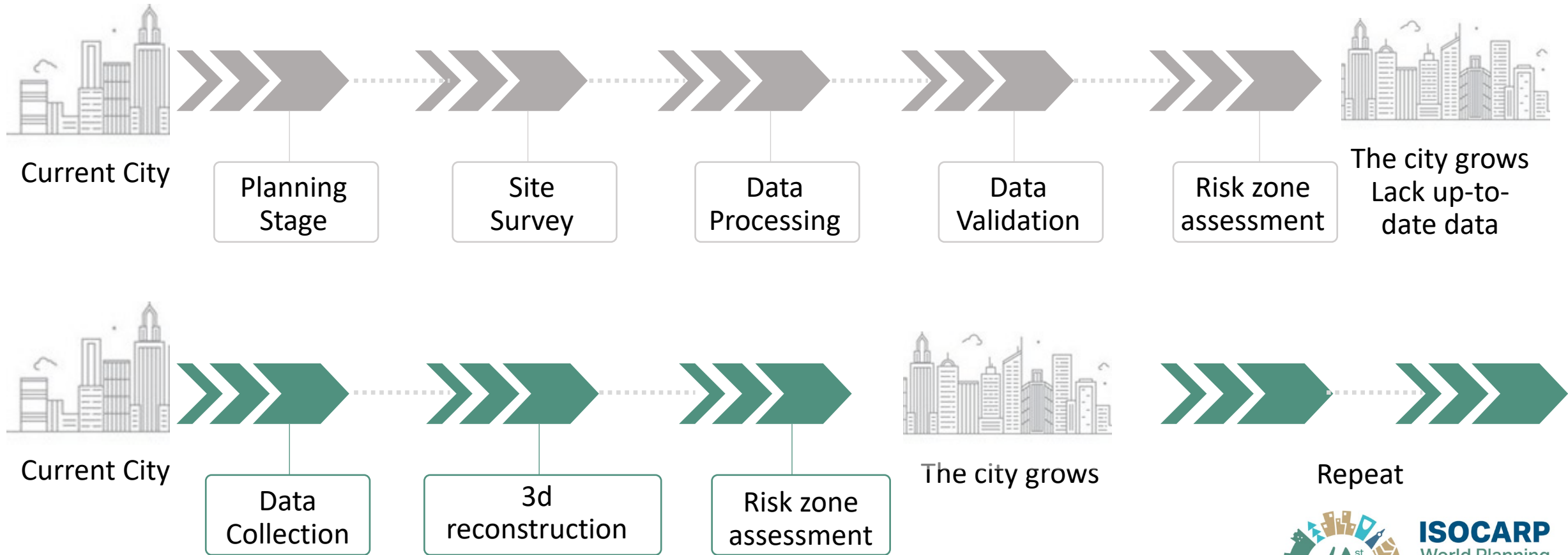
Available at : (https://www.google.com/maps/@51.05404,3.72405,190m/data=!3m1!1e3?entry=ttu&g_ep=EgoyMDI1MTAwNi4wIKXMDSoASAFQAw%3D%3D)
(Accessed: 8 October 2025).

Digital Urban Twins (n.d.) [Flanders region, Belgium, DTC of Flanders region, Belgium] [Online image].

Available at : (<https://www.digitalurbantwins.com/flanderstwin>)(<https://www.digitalurbantwins.com/flanderstwin>) (Accessed: 8 October 2025).

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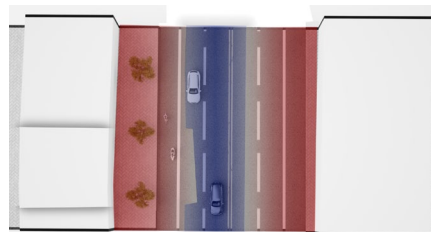
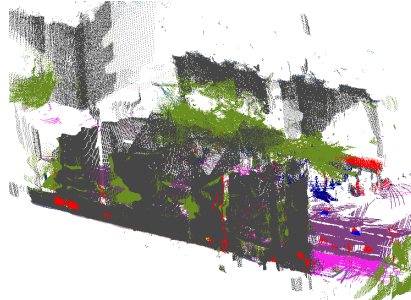
Explaining and Defining the Research – Keeping pace with urban development



Explaining and Defining the Research – Urban resilience through accessibility



City lack of safety data



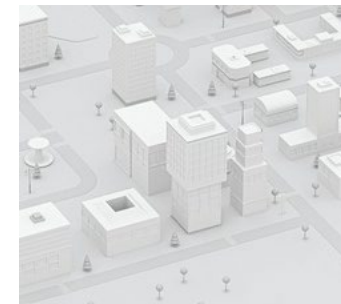
Risk Zone mapping



Quality of life

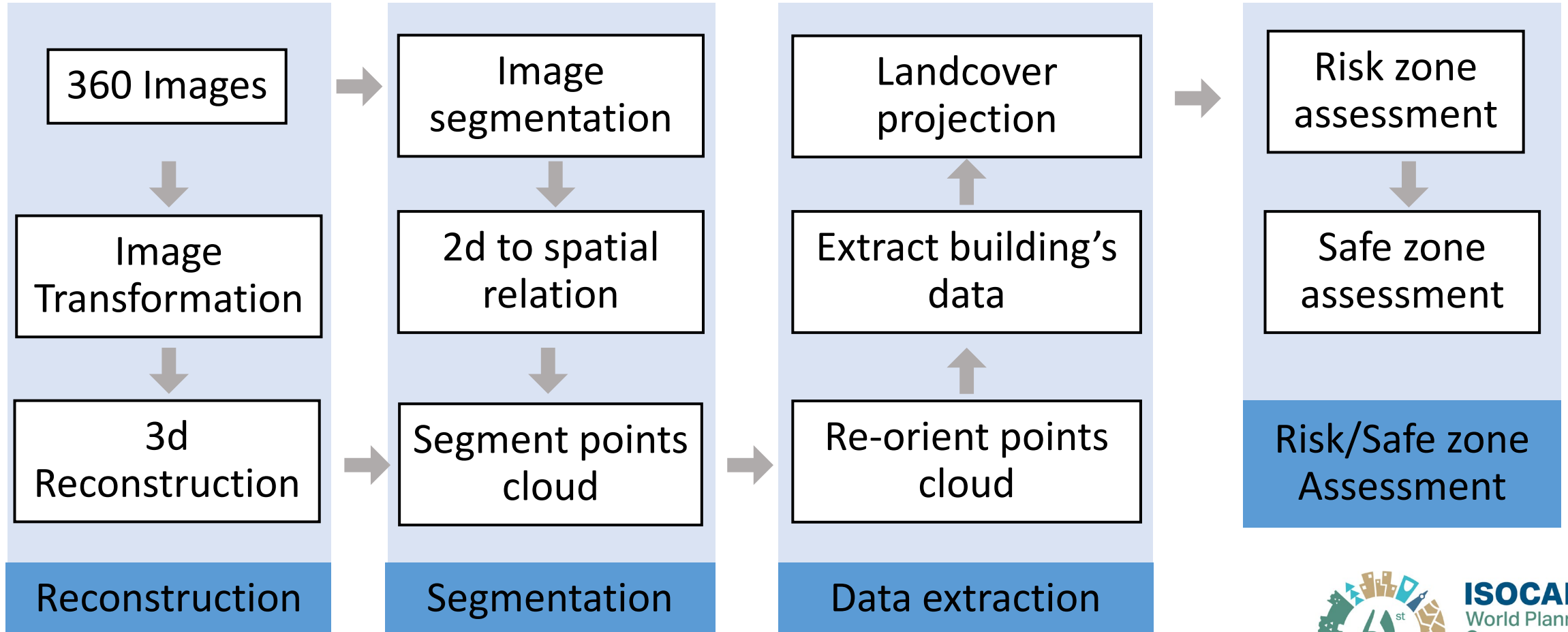


Resilience city

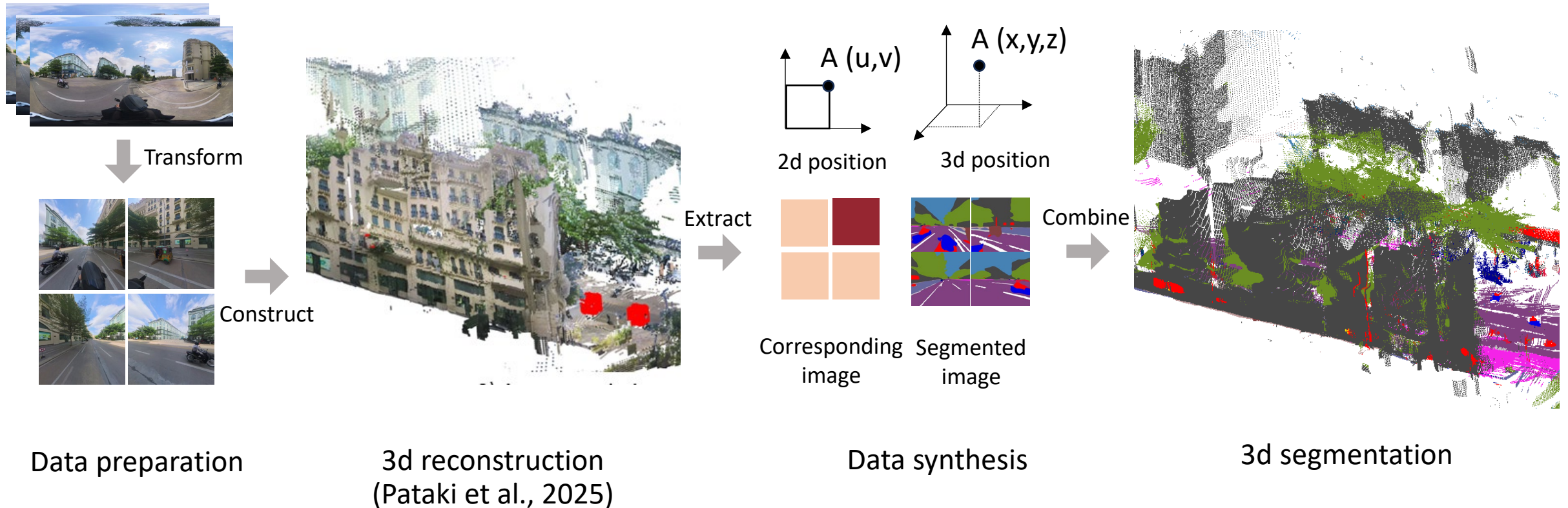


Affordability

Methodology - Framework Overview



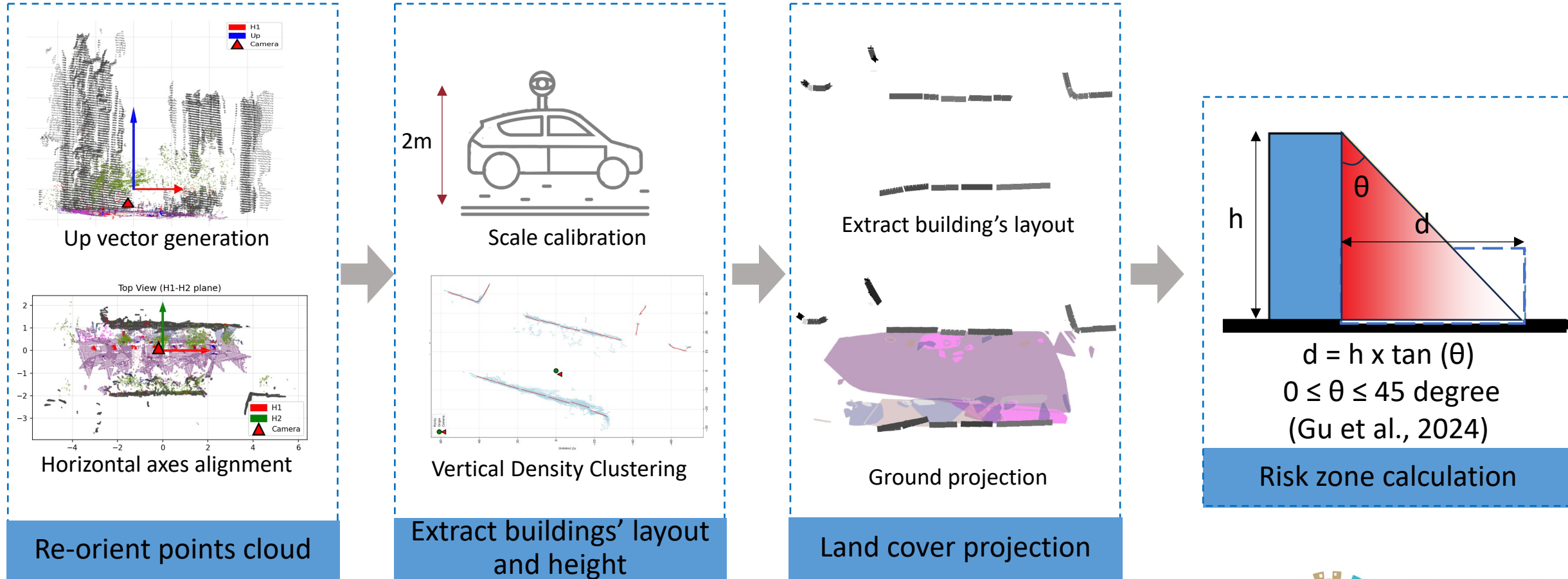
Methodology - 3d segmentation



Pataki, Z., Sarlin, P. E., Schönberger, J. L., & Pollefeys, M. (2025). MP-SfM: Monocular Surface Priors for Robust Structure-from-Motion. In Proceedings of the Computer Vision and Pattern Recognition Conference, pp. 21891-21901.

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Methodology – 3d reconstruction



Gu, Z., Sun, X., & Zhu, H. (2024). A simple method of measuring the seismic risk of circulation space in urban residential streets based on street-view data. Paper presented at the Annual Meeting of the Geographic Information Systems Conference, October 2024.

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Study Results



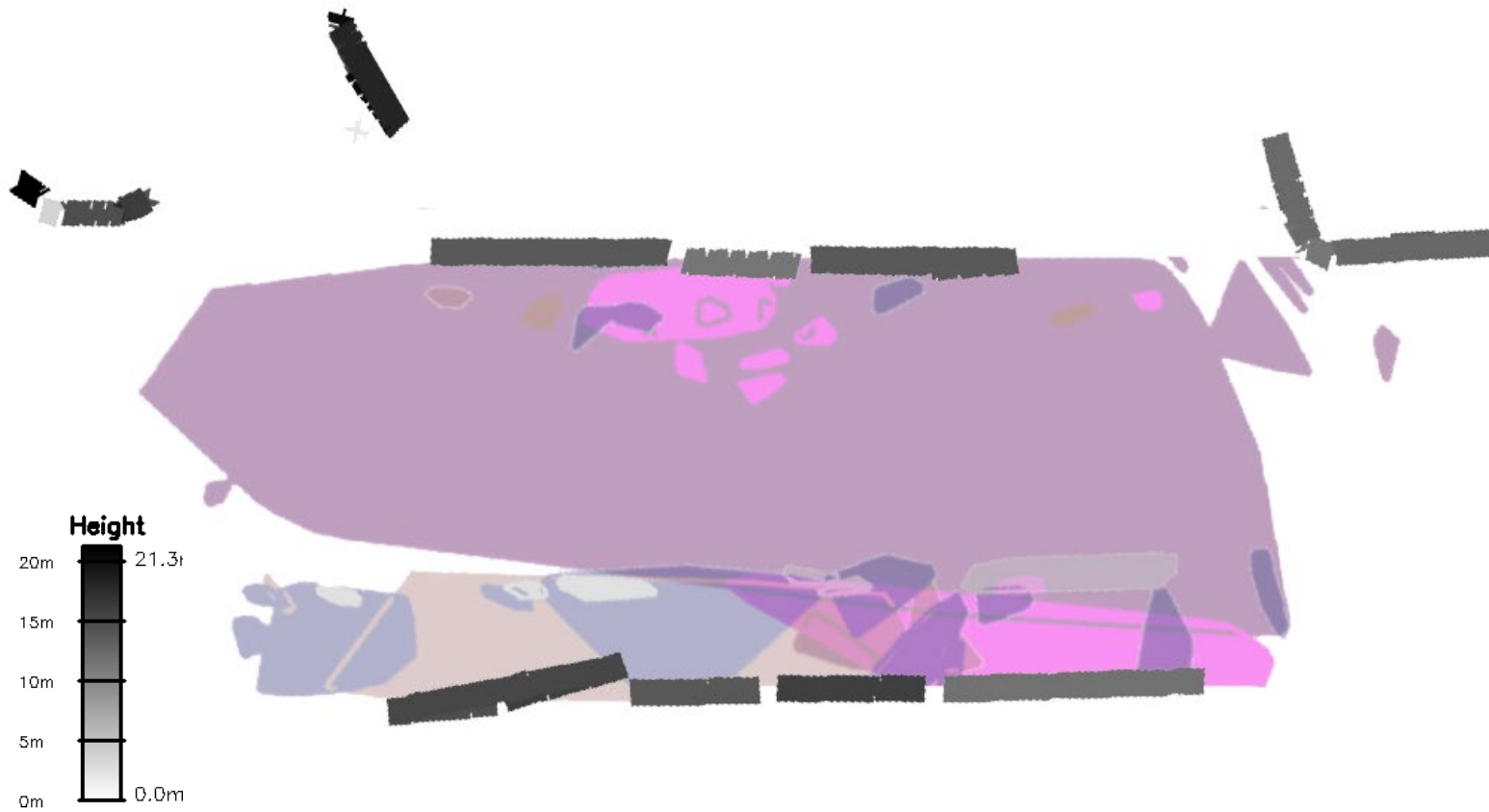
Parameters	Properties
Distance	10 meters
Camera	Insta360 X3
Resolution	5760 x 2880 pixels
Quality	96 dpi
Road Segment's Length	70 meters
Location	Koh Pich, Phnom Penh, Cambodia

Data parameters

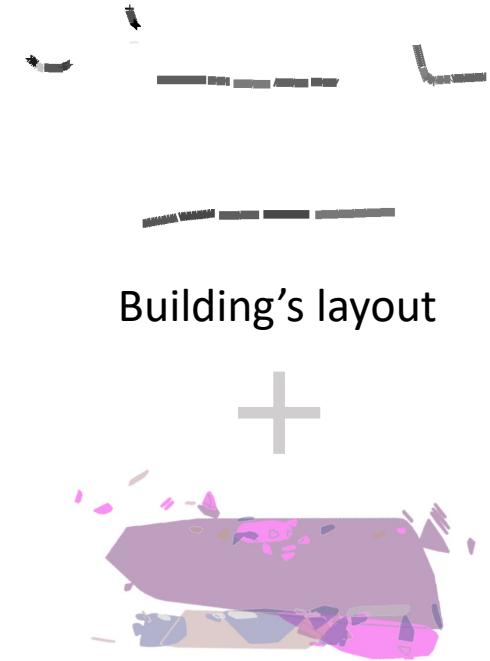


Case study location

Study Results



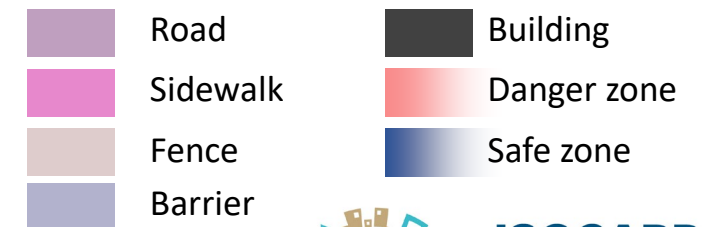
Projected landcover with the buildings' layout



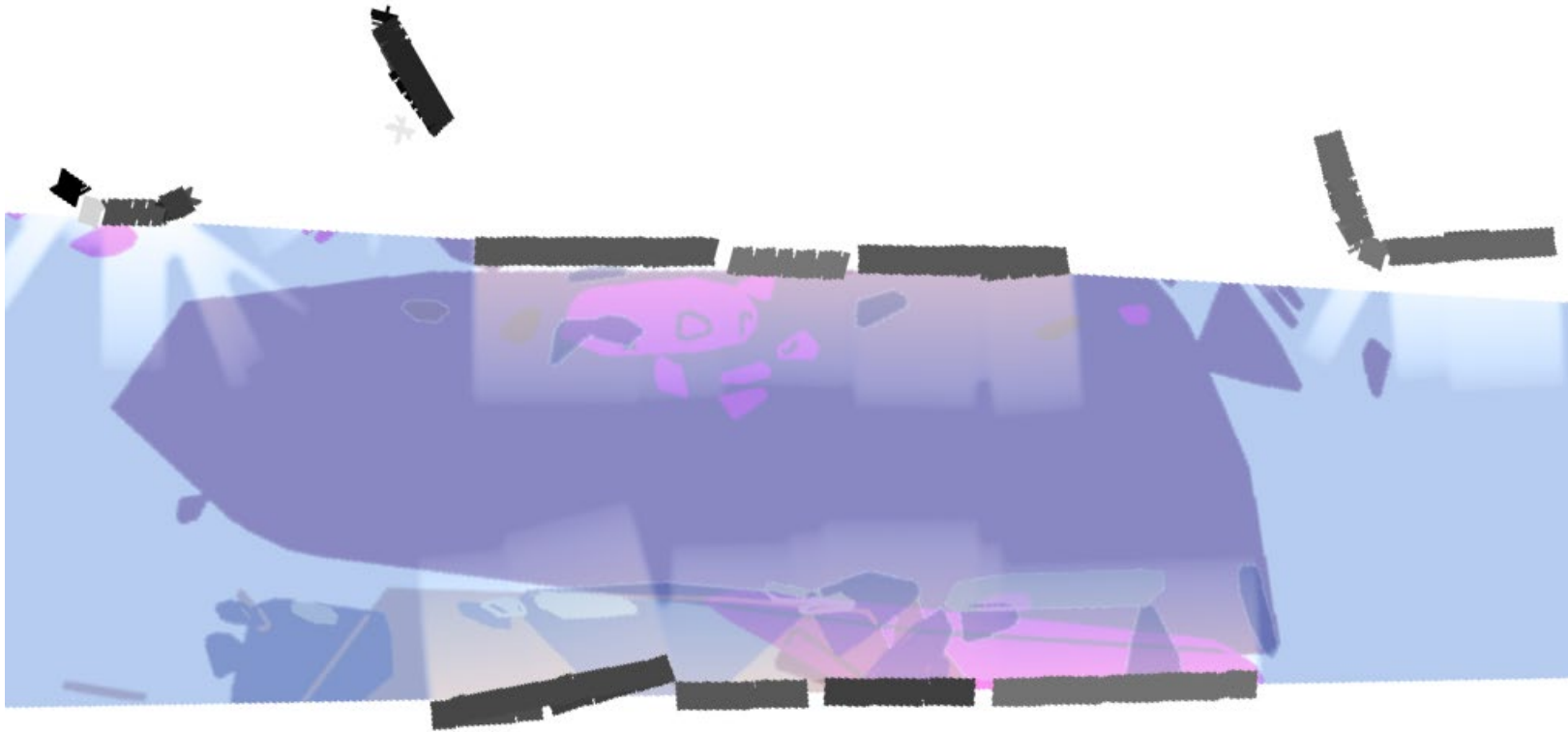
Building's layout



Ground projection



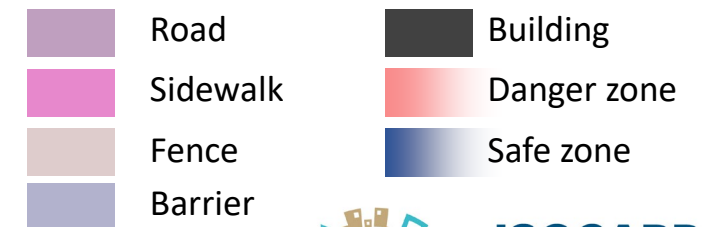
Study Results



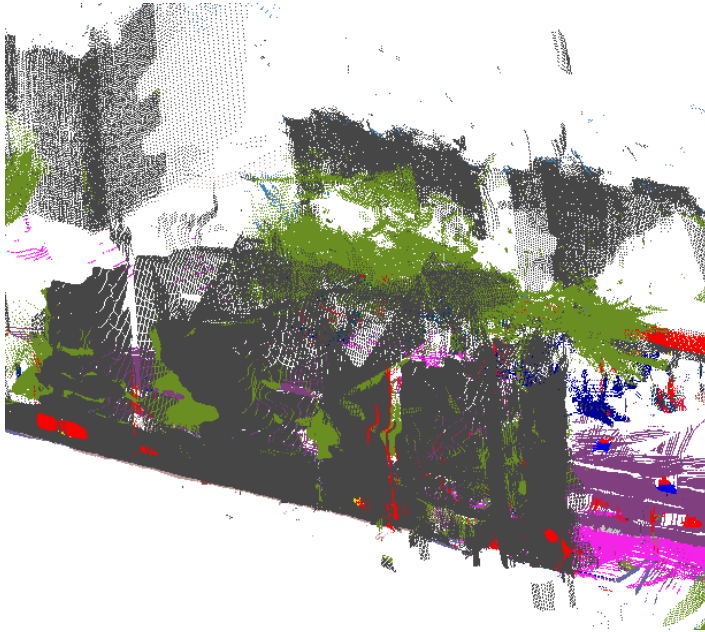
Safe zone assessment



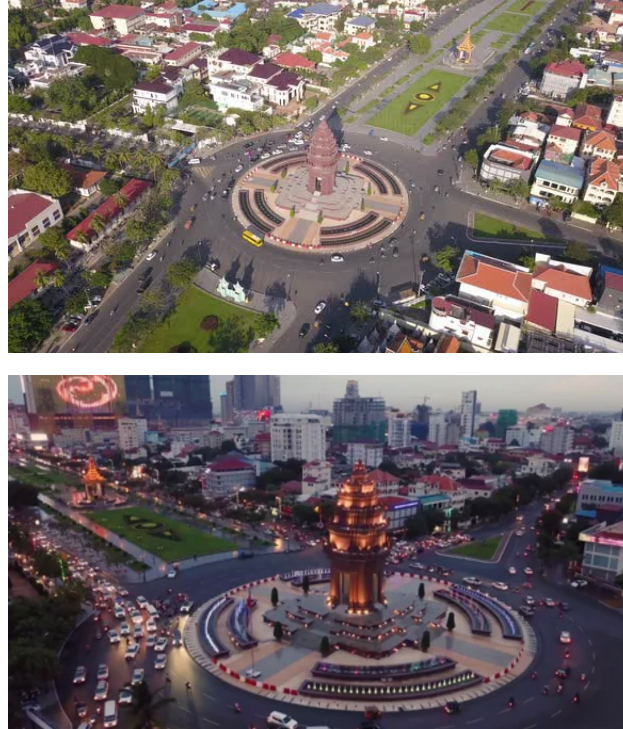
Risk zone assessment



Broader Study Impact – Spatiotemporal analysis



Reconstructed space

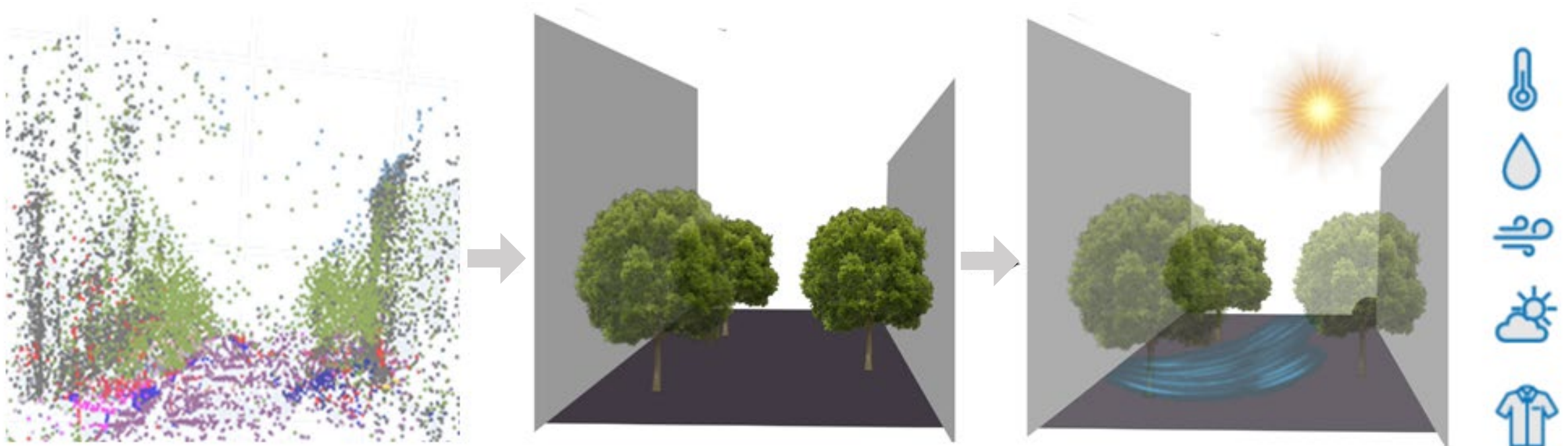


Time iteration



Behavioural/
Event reconstruction study

Future Development – Thermal comfort simulation

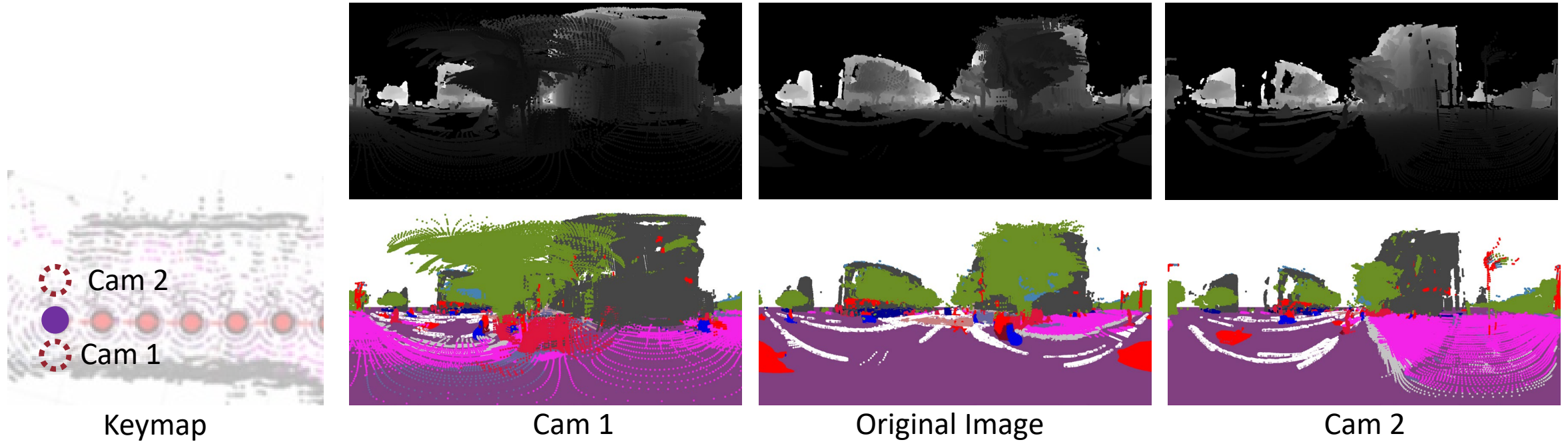


Segmented
points cloud

3d mesh generation

Thermal comfort simulation

Future Development – Iterating camera's pose



Summary – Contribution

- Segmented 3d points cloud from 360 images
- Extracted building's layout from points cloud data
- Estimated debris fall area using 360 images



Summary – Limitation

- The reconstructed points cloud may incomplete data
- The workflow assumes the ground is flat
- The height of the building is calibrated by assuming the camera's height is 2m from the ground

